

HARP: ExactFrac-Constrained Score Serving for GNN Prediction APIs

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Abstract—GNN prediction APIs emit softmax scores consumed by ranking, routing, A/B tests, and compliance pipelines. Production serving systems cache predictions by request hash and assume deterministic outputs for cache hits and audit replay; randomized release noise breaks those assumptions. We formalize the resulting operational requirement as ExactFrac—the fraction of responses that remain bitwise identical under re-query—and study Acc under $\text{ExactFrac} \geq c$ (study default $c=0.60$; c is an operator dial). Any fresh uniform posterior perturbation (e.g., LBP) yields $\text{ExactFrac}=0$ almost surely; session-sticky noise is not a substitute (cross-client ExactFrac and pre-session audit replay stay 0). We propose HARP, a constrained score-serving framework: constructors rank nodes, hop expansion forms a protected set, a pluggable protector runs only there, and CFS maximizes Acc under $\text{ExactFrac} \geq c$ and an optional audit cap matched to the attack budget. Selective release is post-processing, so HARP composes with any trainer—including DP trainers such as GAP—without degrading (ϵ, δ) . Under defense-aware LiRA at $n_{\text{shadows}}=16$ (5 seeds), HARP is feasible for $c>0$ while uniform perturbation is not, with competitive Acc/ECE on a bit-stable clean majority. Constructor choice is a measured slice-privacy dial ($0.669 \rightarrow 0.594$, $p<10^{-3}$); residual clean-slice risk is a structural limit of $\text{ExactFrac}>0$.

Index Terms—graph neural networks, prediction APIs, ExactFrac, selective release, score serving, response caching

I. INTRODUCTION

A. Motivation

GNN prediction APIs return softmax vectors for triage, ranking, and analytics [9], [10]. At scale the API is a continuous data product with Volume (graphs from 10^3 to 10^6 nodes), Velocity (clients re-query), and Veracity (downstream systems need stable, well-calibrated scores). Released scores also leak training membership [1], [2], [4], [5].

a) Why ExactFrac (operational, not invented).: Prediction serving is built around *deterministic* request→response maps: Clipper caches predictions to cut latency under repeated queries [40], and NVIDIA Triton hashes model+inputs to return cached outputs without re-execution [41]. Those mechanisms are correct only when identical inputs yield identical outputs; a fresh Laplace draw on every query makes the cache key a moving target and breaks audit replay of logged decisions. Score-threshold routers (route if $p_{\text{top}}>\theta$) likewise need non-flickering scores under re-query. We *formalize* that operational need as ExactFrac—the fraction of

node responses bitwise identical under re-query—and study Acc subject to $\text{ExactFrac} \geq c$. Here $c=0.60$ is a locked *study default*, not a claimed industry constant; Fig. 2 sweeps $c \in \{0.2, 0.4, 0.6, 0.8\}$ as an operator dial. Fresh uniform LBP cannot meet any $c>0$ (Remark 1). Session-sticky LBP ($B=1$) is the obvious alternative: Table VIII shows it restores some per-client cache hits but keeps cross-client $\text{ExactFrac} \approx 0$, fails pre-session audit replay, and leaves clean-slice threshold flicker—so sticky noise is not ExactFrac.

b) Systems problem.

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{ExactFrac}(\Pi) \geq c, \quad \text{LiRA}(\Pi) \leq \tau$$

when the second constraint is active; otherwise maximize Acc subject to ExactFrac alone. Uniform mechanisms are infeasible for $c>0$.

c) Framework.: **HARP** is a constrained score-serving framework with three interfaces: (1) a *constructor* that ranks nodes (topology, degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability); (2) hop expansion $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$ plus a *protector* applied only on $\mathcal{P}_k$; (3) operator dials Frac/Mass/session budget B , with **CFS** searching Frac under (c, τ) . Default configuration is *release-only* selective Laplace at $\text{Frac}=0.40$ ($\text{ExactFrac}=0.60$).

d) Headline evidence.: At $n_{\text{shadows}}=16$, HARP meets $\text{ExactFrac} \geq 0.60$ while uniform LBP cannot; equal-mass LBP remains the right comparator when $c=0$ is allowed. Within the constrained class, constructors cut clean-slice AUROC ($0.669 \rightarrow 0.594$), CFS is grid-optimal, and HARPoGAP preserves GAP’s ϵ . Measured replay/flicker and multi-tenant cache stress validate ExactFrac as an operational (not only definitional) property.

B. Gap

Prior release defenses (LBP [5], MemGuard [6]) perturb every response and are incompatible with prediction caches that assume deterministic outputs [40], [41]. Train-time GNN MIA methods (GTD, MaskArmor) [21], [22] and DP-GNNs [13], [14], [36] do not expose ExactFrac/Mass serving accounting. None formalize $\text{ExactFrac} \geq c>0$ as a constrained release objective with constructors, CFS, and trainer composition.

a) *Running example* (Cora, $c=0.60$, $n_{sh}=16$).:

Equal-mass LBP: $\text{Acc} \approx 0.87/\text{LiRA} \approx 0.59/\text{ExactFrac} = 0$.
HARP: $\text{Acc} \approx 0.83/\text{LiRA} \approx 0.67/\text{ExactFrac} = 0.60$ with much lower clean-majority ECE. Strong LBP collapses Acc at $\text{ExactFrac} = 0$ —secondary when $c > 0$.

C. Contributions

- 1) **ExactFrac-constrained release.** Formalize Acc s.t. $\text{ExactFrac} \geq c$ (optional $\text{LiRA} \leq \tau$ matched to the audit budget); show uniform perturbation is infeasible for $c > 0$; give a hop-coverage argument, CFS grid optimality (Prop. 2), and DP post-processing composition (Prop. 3).
- 2) **HARP serving layer.** Trainer-/protector-agnostic selective release with constructors that lower clean-slice AUROC, CFS under the headline LiRA budget, optional DCS, and HARPoGAP for finite ε plus ExactFrac controls.
- 3) **Constraint-aware evaluation.** Headline LiRA at $n_{sh}=16$; sticky-vs-ExactFrac serving evidence (Table VIII); canary-stressed OGB probe; clean-slice risk reported as a structural limit.

II. BACKGROUND AND RELATED WORK

Shadow/confidence MIAs [1]–[3] and LiRA [4] are the audit standard for prediction APIs; overconfidence is a recurring cue [17]. On graphs, Olatunji et al. [5] introduce node-level MIAs and LBP; He et al. [38], GTD [21], MaskArmor [22], and surveys [8] study related attacks/defenses. MemGuard [6] adversarially perturbs every confidence vector ($\text{ExactFrac}=0$). DP-GNNs such as GAP [13], [14] provide train-time (ε, δ) ; structure modulates leakage [18], [39]. **Positioning:** LBP/MemGuard spend on every node; GTD/MaskArmor harden training without ExactFrac accounting; GAP answers formal ε ($\text{ExactFrac}=1$ at eval for train-only noise); HARP answers $\text{ExactFrac} \geq c$ serving with optional audits and composes with any trainer. Our GAP-agg and MemGuard baselines are faithful reimplementations, not line-by-line ports of the official codebases—we label them as such throughout. Primary comparisons under $c > 0$ are selective policies at matched Frac; equal-mass LBP is the unconstrained ($c=0$) reference.

III. PROBLEM SETUP

A. Notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ nodes, feature matrix $X \in \mathbb{R}^{n \times d}$, and labels $y \in \{0, \dots, C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{Q}$ with ratios 40%/20%/40% (resampled per seed). Training is *transductive*: every node participates in message passing; only $\mathcal{T}$ enters the cross-entropy loss. We write $\hat{p}_v$ for the softmax posterior at v , r_v for a risk score in $[0, 1]$, $\mathcal{P}_k$ for the hop-expanded protected set, $\text{Mass} = \sum_v \sigma_v$, and $\text{Frac} = |\mathcal{P}_k|/n$.

B. Constrained score release

Operators rarely optimize Acc alone. They need usable accuracy and a reproducible majority of API responses, subject to a membership audit.

Definition 1 (Constrained Score Release). *Given a trained GNN and release channel, choose a release policy Π to*

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{LiRA}(\Pi) \leq \tau, \quad \text{ExactFrac}(\Pi) \geq c,$$

where *ExactFrac* is the fraction of nodes whose released posterior is bitwise identical under independent re-query (equivalently equals $\hat{p}_v$ under release-only selective Laplace with $\text{weak}=0$), τ is an audit threshold matched to the LiRA budget, and $c \in [0, 1]$ is an operator-chosen clean fraction (study default 0.60).

Remark 1 (Uniform LBP is infeasible for $c > 0$). *Independent Laplace noise with scale $b > 0$ on every coordinate changes the value almost surely, so $\text{ExactFrac} = 0$ for any uniform LBP policy. Selective release with protected fraction Frac achieves $\text{ExactFrac} = 1 - \text{Frac}$ by construction.*

This reframes the Acc–LiRA comparison. Equal-mass LBP can win Acc subject to $\text{LiRA} \leq \tau$ when $c=0$ is allowed; HARP solves the strictly harder problem with $c > 0$. The Frac sweep traces HARP’s Pareto frontier under that constraint.

C. Threat model

The adversary is a black-box client: it queries softmax scores and tries to guess whether a node was in the training set [2], [4], [5]. It does not see gradients or embeddings. White-box parameter access remains out of scope; label-only gap AUROC and edge-similarity membership AUROC are reported alongside LiRA (GAP-agg cuts edge AUROC $0.895 \rightarrow 0.748$ on Cora).

We also stress three practical variants: defense-aware shadows, multi-query averaging ($K \in \{1, 5, 20\}$), and an LTE-aware query adversary that focuses on clean nodes, protected-set boundaries, and top-decile LTE seeds (Sec. VI-B0l). We do not claim robustness to poisoning or unbounded queries without a session budget.

D. Metrics

Utility: test Acc (primary). **Audit:** LiRA AUROC [4] and $\text{TPR}@1\% \text{ FPR}$, plus Salem-style ϕ /confidence attacks [2]. **Fidelity:** ECE [17], clean-slice ECE, ExactFrac. LiRA decides *whether* to spend noise; Acc/ECE measure the cost.

Remark 2. *The primary protocol is transductive (non-members remain in the graph during training), matching common semi-supervised GNN APIs. We additionally report an inductive edge-exclusion check on Cora (Sec. VI-B0m) where test-incident edges are dropped at train/infer time.*

IV. HARP: HOP-AWARE SELECTIVE RELEASE

A. Design principle

Uniform posterior noise cannot meet $\text{ExactFrac} \geq c > 0$ —any fresh $b > 0$ draw changes every response. Node-local selection



Fig. 1. HARP pipeline: risk ranking $\rightarrow$ high-risk seeds $\rightarrow$ k -hop expansion $\rightarrow$ protector only on the protected set.

without hop coverage is incomplete for GNNs: if v is protected but neighbor u is released exact, aggregation can reintroduce membership signal into u 's score. HARP is a three-interface serving framework (Fig. 1):

- 1) **Constructor (pluggable)**: rank nodes—topology (LTE), degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability.
- 2) **Geometry + protector**: expand $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$; apply Laplace or masking only on $\mathcal{P}_k$.
- 3) **CFS + dials**: search $\text{Frac} \geq c$ and optional $\text{LiRA} \leq \tau$; $\log \text{Mass}/\text{Frac}/B$.

Locked study config: *release-only* Laplace, $\text{Frac}=0.40$ ($\text{ExactFrac}=0.60$), $k=1$, $\sigma=0.30$. Topology LTE is the no-shadow default constructor; audit-guided ranking is preferred when shadows are already paid for. Train-time alignment is an optional ablation—not required for the ExactFrac claim.

a) *Noise mass and clean fraction.*: $\text{Mass} = \sum_v \sigma_v$ and $\text{Frac} = |\mathcal{P}_k|/n$ are operator accounting metrics. Uniform LBP at scale b has $\text{Mass}=bn$ and $\text{Frac}=1$. See paragraph IV-E0a for the locked-budget identity.

b) *Locked-config example (Cora).*: On Cora ($n=2708$), $\text{Frac}=0.40$ protects ≈ 1083 nodes after one-hop expansion; ≈ 1625 nodes stay exact. $\text{Mass}_{\text{HARP}} \approx 325$ matches equal-mass LBP with $b=0.12$. Strong LBP ($b=0.3$) spends $\text{Mass} \approx 812$ on all nodes—the Acc-collapse baseline in Table II.

c) *Release transform (Laplace protector).*: Let $\sigma_v = \sigma_{\text{strong}} \mathbb{1}[v \in \mathcal{P}_k]$. The released posterior is

$$\tilde{p}_v = \text{renorm}([\hat{p}_v + \text{Lap}(0, \sigma_v)]_+), \quad (1)$$

with independent Laplace coordinates and the same transform on every defense-aware shadow. The masking protector instead returns $\hat{p}_v$ for $v \notin \mathcal{P}_k$ and a one-hot of $\arg \max \hat{p}_v$ for $v \in \mathcal{P}_k$.

d) *Algorithm (systems view).*:

- 1) **Risk**: $r \leftarrow \text{Constructor}(G, \mathcal{T})$ (topology / degree / confidence / random / audit).
- 2) **CFS (optional)**: search $\text{Frac} \leq 1 - c$ to maximize Acc subject to $\text{LiRA} \leq \tau$.
- 3) **Seeds + expand**: binary-search seeds so $|\text{Ball}_k(\mathcal{R})|/n \approx \text{Frac}$; $\mathcal{P}_k \leftarrow \text{Ball}_k(\mathcal{R})$.
- 4) **Release**: apply (1) (or masking); $\log(\text{Mass}, \text{Frac}, \text{Acc}, \text{LiRA}, \text{ECE})$.

Train-time alignment is optional and omitted in the locked release-only configuration.

B. Risk constructors

The constructor interface is part of the framework contribution. **Topology (LTE)** uses only $G, y|_{\mathcal{T}}$, and the train mask (inverse degree, local heterophily, train-neighbor exposure); it is a cheap default when shadows are unavailable—not claimed to be privacy-optimal. **Degree / train-neighbor** isolate individual topology cues. **Confidence / entropy** rank from a clean forward pass (high max-confidence or high entropy). **Random** is the matched-Frac selectivity control. **Audit-guided** scores $|\mu_{\text{IN}} - \mu_{\text{OUT}}|$ over a small shadow ensemble [37] and is preferred when that ensemble is already paid for. We report Spearman correlation between each constructor and audit vulnerability, plus Acc/LiRA/slice AUROC at matched Frac (Table VII). On Cora/Citeseer, *entropy* is the strongest cheap prior (Spearman ≈ 0.33 – 0.35); topology (LTE) is moderate (≈ 0.25 – 0.28); max-confidence is anti-correlated. Heterophilic Chameleon priors are weak—selectivity still enforces Exact-Frac, but ranking gains are limited. Where topology underperforms random on LiRA (e.g., Cora in Table VII), we state that plainly: ExactFrac comes from selectivity, not from LTE.

C. Why k -hop expansion

Let $\mathcal{R}$ be seeds. A GNN score at u already mixes information from $\text{Ball}_k(u)$ before any release noise is applied. If a membership-sensitive seed $v \in \mathcal{R}$ is noised at release but a neighbor u is returned bit-exact, the adversary simply queries u .

Proposition 1 (Hop coverage under a one-hop linear mixer). *Fix a one-hop score model $s_u = \alpha z_v + (1 - \alpha) z_u$ for $u \in \mathcal{N}(v)$, $\alpha \in (0, 1]$, where z_v carries a membership cue of gap $\gamma > 0$, and release adds independent Laplace noise only to chosen nodes' scores. If the policy noises v but leaves u exact, an adversary observing s_u retains the unprotected leaf advantage. Noising $\text{Ball}_1(\{v\})$ places Laplace noise on every score that linearly depends on z_v under this mixer.*

Proof sketch. Under seed-only protection the released leaf equals the clean mixer output, so the member/non-member gap in $\mathbb{E}[s_u]$ remains $\alpha\gamma$. Expanding protection to u subjects that coordinate to $\text{Lap}(0, \sigma)$. A synthetic mixer confirms the ordering: unprotected and seed-only leaf AUROCs stay near 1.0, while hop-1 protection lowers the leaf attack to ≈ 0.72 . $\square$

This is a motivating linearized argument for hop expansion, not a necessity theorem for nonlinear multi-layer GNNs. Empirically we set k to the model's receptive field (default $k=1$ for 2-layer SAGE/GCN; $k \in \{0, 2\}$ changes Acc by $\lesssim 0.01$ at matched Frac). Frac—not k —is the population Acc–Mass knob.

D. Guarantees: CFS optimality and DP composition

Proposition 2 (CFS grid optimality). *Let $\mathcal{F} = \{f_1 < \dots < f_T\} \subseteq [0, 1 - c]$ be the CFS grid with spacing δ , and let $\text{Acc}(f)$, $\text{LiRA}(f)$ be the (validation-estimated) accuracy and audit curves over Frac f . CFS returns $f^* =$*

$\arg \max \{ \text{Acc}(f_i) : \text{LiRA}(f_i) \leq \tau \}$, the exact constrained optimum over the finite grid $\mathcal{F}$ (enumeration, not a continuous convex program). If additionally Acc is L -Lipschitz and LiRA is nonincreasing in f , then $\text{Acc}(f^*) \geq \sup \{ \text{Acc}(f) : f \in [0, 1 - c], \text{LiRA}(f) \leq \tau \} - L\delta$.

Proof sketch. Exactness over $\mathcal{F}$ is by enumeration. For the continuous bound, monotone LiRA makes the feasible set an interval containing a grid point within δ of any continuous optimum; Lipschitz Acc loses at most $L\delta$. Empirically Acc falls ≈ 0.02 per 0.1 Frac (Table IV), so $L\delta \approx 0.004$ at $\delta = 0.02$. $\square$

Proposition 3 (Post-processing composition; trainer-agnostic). *Let T be any training procedure whose released model satisfies (ε, δ) -DP w.r.t. the training data, and let Π be any HARP release policy applied to the model's outputs. Then $\text{HARP} \circ T$ satisfies the same (ε, δ) -DP (standard post-processing [27]) and attains $\text{ExactFrac} = 1 - \text{Frac}$ w.r.t. the DP model's posteriors.*

Proof. Π reads only model outputs plus public graph structure. $\square$

Prop. 3 is not a new DP theorem; it is the reason HARP can sit *above* any trainer (DP or not) as a serving layer. Section VI-B measures $\text{HARP} \circ \text{GAP}$.

E. Complexity and accounting

Proposition 4 (HARP setup and release complexity). *Let $G = (V, E)$ have $n = |V|$ nodes and $m = |E|$ adjacency endpoints, with C classes. Fix hop radius $k = O(1)$ and $T = O(1)$ binary-search iterations for a target protected fraction. Then setup (LTE + seeds + k -hop expand) runs in $O(m + n \log n)$ time and $O(n + m)$ space. Release runs in $O(nC)$ —matching uniform LBP's tier, with less noise-sampling work whenever $\text{Frac} < 1$.*

Proof sketch. LTE scans edges then normalizes n scores: $O(m + n)$. Seed selection is $O(n \log n)$ plus T BFS expansions at $O(m + n)$. Release writes scales, samples Laplace on $\mathcal{P}_k$, and renormalizes: $O(nC)$. $\square$

Remark 3 (Per-response Laplace; no global DP claim). *Interpret each node's clean posterior as a query with ℓ_1 -sensitivity at most $\Delta = 2$ on the simplex. Independent $\text{Lap}(\sigma)$ on coordinates of $v \in \mathcal{P}_k$ is a per-response Laplace mechanism with scale σ ; unprotected nodes receive no noise. Any policy with $\text{ExactFrac} > 0$ has infinite global (ε, δ) -DP by construction. We report this only as an accounting remark—not as a local-DP guarantee under graph adjacency—and recommend training-time DP (e.g., GAP) when a legal global ε is required [13].*

a) *Accounting identity (stated once).*: Under locked Frac and fixed σ_{strong} , $\text{Mass}_{\text{HARP}} = \text{Frac} \cdot n \cdot \sigma_{\text{strong}}$ by construction. Mass is an operational dial correlated with mean ℓ_1 distortion and top-1 flip rate (at matched Mass 325 on Cora: HARP mean ℓ_1 0.349 vs. LBP 0.493); it is not a standard privacy budget. Empirical claims are (i) feasibility

under $\text{ExactFrac} \geq c$, (ii) Acc/ECE within the selective class at matched Frac/Mass , and (iii) serving telemetry.

F. What HARP is not

HARP is not differential privacy and is not a claim of lower population LiRA than uniform noise under $\text{ExactFrac} = 0$. It is not a new Laplace mechanism. It is a constrained ExactFrac serving framework: spend a protector on a hop-expanded subset, keep the rest replay-stable, search Frac with CFS, and audit residual risk—including on unprotected nodes.

V. EXPERIMENTAL PROTOCOL

A. Pre-declared rules

Primary objective: Acc under $\text{ExactFrac} \geq c$ (study default $c = 0.60$; swept in Fig. 2). **Primary metrics:** Acc , ExactFrac , ECE (fidelity). **Audit metric:** LiRA AUROC (and clean-slice confidence AUROC) as deployment constraints—not the headline win condition. **Operational metrics:** Mass , Frac , cross-client ExactFrac , audit replay, shared-cache hit rate / QPS (Table VIII). Hyperparameters are not selected to minimize test LiRA . Under $c > 0$, primary baselines are selective policies at matched Frac ; equal-mass LBP is the unconstrained ($c = 0$) reference; sticky LBP is the false substitute for ExactFrac .

B. Systems metrics (definitions)

- **ExactFrac:** fraction of nodes whose released posterior equals $\hat{p}_v$ bit-exactly.
- **Mass** = $\sum_v \sigma_v$: operational Laplace-scale spend (correlated with distortion).
- **Frac** = $|\mathcal{P}_k|/n$: protected fraction ($\text{ExactFrac} = 1 - \text{Frac}$ under weak = 0).
- **ECE:** calibration of released probabilities.

C. Locked HARP (release-only)

Release-only selective Laplace: $\lambda = 0$, no HCAG, no train-on-protected; $k = 1$, $\sigma_{\text{strong}} = 0.30$, $\text{Frac} = 0.40$ ($\text{ExactFrac} = 0.60$). Constructor default in locked tables: topology (LTE); audit/random/degree/confidence appear in Table VII. Train-time alignment is reported only as an ablation. Baselines: equal-mass LBP $b = 0.12$; strong LBP $b = 0.3$; MemGuard; GTD; MaskArmor; selective masking; DP-SGD Acc -tuned (vacuous ε).

D. Baselines and attacks

Gaussian LiRA with defense-aware shadows: $n_{\text{shadows}} = 16$ for the headline Cora table and CFS (Tables II, V; $\tau = 0.70$), $n_{\text{shadows}} = 4$ for multi-dataset grids and Acc dials, and sweeps $\{4, 16, 64\}$ on Cora/Chameleon. Five seeds with mean $\pm 95\%$ CI on primary Cora/Chameleon/Citeseer constructor tables; paired seeds across defenses. ogbn-arxiv: NeighborLoader serving Acc with LiRA at $n_{\text{shadows}} \in \{2, 4, 16\}$ (audit stress, not a million-node privacy claim). ogbn-products: full-graph serving RSS/QPS ; BFS LiRA ladder at $n \in \{15k, 40k\}$, $n_{\text{sh}} \in \{2, 4, 8\}$, plus a canary-stressed 15k probe with unfenced canary- $\text{LiRA} \approx 1.0$.

E. What each baseline tests

Equal-mass LBP ($b=0.12$) matches HARP Mass at $\text{Frac}=0.40$ and is the unconstrained ($c=0$) reference. **Selective constructors** (random/degree/confidence/entropy/audit/topology) isolate ranking quality at matched Frac under the SLA. **Strong LBP** ($b=0.3$) is a secondary published recipe that collapses Acc under full-graph strong noise. **MemGuard** (reimplementation) fits a standardized MLP ϕ -attack on holdout members/non-members and searches an L_1 ball to drive $P(\text{member}) \rightarrow 0.5$ while preserving argmax ($\text{ExactFrac}=0$). Following Aerni et al. [37] we report both ϕ and LiRA; at $n_{\text{sh}}=16$ our port cuts LiRA vs. undefended ($0.813 \rightarrow 0.667$) while remaining ExactFrac -infeasible. **GAP-agg** is a GAP-inspired aggregation-Laplace baseline: noise is train-only, so eval $\text{ExactFrac}=1$ with analytical (ε, δ) (hybrid DP-train + ExactFrac -serve). Not a full GAP codebase reproduction. **Selective masking** is an alternate protector (ExactFrac preserved; often worse LiRA). **HARP-full** ($\text{Frac}=1$) is the unconstrained ablation that recovers strong-LBP Acc-LiRA. **GTD/MaskArmor** are train-time defenses (no release Mass/ Frac). **Clip+noise DP-SGD** remains an Acc-tuned vacuous- ε reference.

F. Attack protocol (summary)

We train defense-aware shadow models, release transformed posteriors, and report LiRA AUROC with ϕ -features [2], [4]. Validation nodes are excluded from membership labels; test Acc/ECE use the held-out test split. Multi-query experiments average $K \in \{1, 5, 20\}$ independent draws; session policies cap fresh draws at $B=5$.

a) *Negative controls.*: PubMed (near-chance undefended LiRA): HARP still preserves Acc vs. strong LBP at $\text{Frac}=0.40$ (0.839 vs. 0.768); ogbn-arxiv/products give near-chance or memory-bounded audits under matched budgets—audit first, harden second [28].

VI. RESULTS

A. Overview

a) *Operational headline.*: Under $\text{ExactFrac} \geq 0.60$: (i) HARP is feasible and uniform LBP is not (Table I); (ii) release-only HARP keeps Acc competitive within the selective class while returning a 60% replay-stable majority; (iii) serving benchmarks show content-cache hits from ExactFrac under fresh traffic (Table IX); (iv) equal-mass LBP can still win Acc-LiRA when $c=0$ is allowed—that unconstrained regime is out of scope for the SLA.

Questions: (Q1) feasibility under $c>0$; (Q2) Acc/ECE vs. selective and equal-mass baselines; (Q3) do constructors beat random on utility and slice privacy; (Q4) serving cost and clean-slice residual risk; (Q5) OGB scale with scoped and leakage-stressed audits. Read the trade space constraint-first: if $\text{ExactFrac} \geq c>0$ is required, uniform rows are infeasible; if only Acc s.t. $\text{LiRA} \leq \tau$ matters, prefer equal-mass LBP.

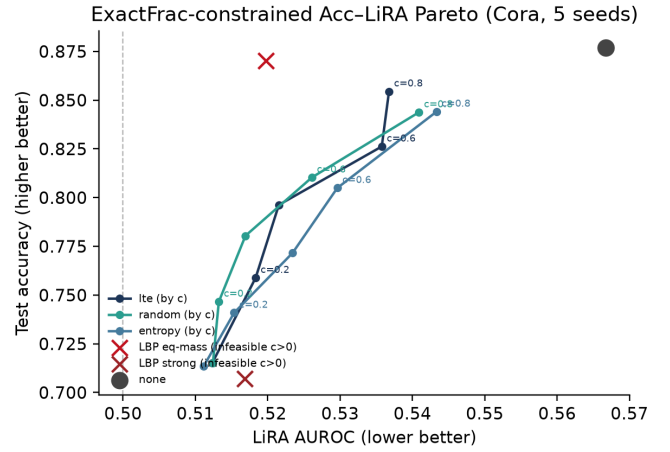


Fig. 2. ExactFrac -constrained Acc-LiRA Pareto on Cora (5 seeds). Curves: selective HARP constructors at $c \in \{0.2, 0.4, 0.6, 0.8\}$ ($\text{ExactFrac}=c$). Uniform LBP markers are $\text{ExactFrac}=0$ and infeasible for every $c>0$.

TABLE I
FEASIBILITY UNDER $\text{ExactFrac} \geq c$. DESIGN $\text{ExactFrac}=1 - \text{Frac}$ IS VERIFIED BY MEASURED RE-QUERY BIT-EQUALITY (TABLE X), NOT LEFT AS A TAUTOLOGY: HARP CLEAN-SLICE FLICKER=0; LBP BIT-EQUAL=0 WITH NONZERO TOP-1/THRESHOLD FLIPS.

Policy	$c=0$	$c=0.2$	$c=0.6$	$c=0.8$
none	yes	yes	yes	yes
LBP (any $b>0$)	yes	no	no	no
LBP + session $B=1$	yes*	no†	no†	no†
HARP ($\text{Frac}=1 - c$)	yes	yes	yes	yes

*Sticky sessions restore cache hits but ExactFrac across clients remains 0.
†Bit-exactness fails for fresh clients and pre-session audit replay.

B. HARP under $\text{ExactFrac} \geq 0.60$: accuracy and score quality

a) *Primary claim.*: Fig. 3, Fig. 4, and Table II: under $\text{ExactFrac} \geq 0.60$, release-only HARP is feasible; equal-mass LBP and MemGuard are not. At $n_{\text{sh}}=16$ the undefended attack is strong (LiRA 0.813, TPR@1% 0.200); HARP cuts it to $0.674 / 0.060$ while keeping Acc 0.826, ECE 0.033, and a 60% replay-stable majority. MemGuard is a credible unconstrained baseline (LiRA 0.667) but $\text{ExactFrac}=0$; equal-mass LBP reaches LiRA 0.592 with ECE 0.159 and $\text{ExactFrac}=0$. Equal-mass can win Acc-LiRA when the SLA is dropped—expected and outside the constrained objective.

b) *Secondary: Acc vs. published strong LBP.*: Against LBP $b=0.3$ [5], HARP raises Acc $0.707 \rightarrow 0.832$ at $\text{Frac}=0.40$ because strong noise applies to a minority of nodes (Table II). This is mechanically expected and is not the constrained-feasibility claim.

c) *Membership audit (deployment dial).*: At matched Mass, equal-mass LBP beats HARP on Acc-LiRA (Cora $n_{\text{sh}}=16$: Acc 0.870 / LiRA 0.592 vs. 0.826 / 0.674). If ExactFrac is not required, prefer equal-mass LBP. Under the SLA, LiRA guides whether Frac must rise and whether the audit threshold τ is met after CFS.

HARP Mass 325; LBP eq-mass 325; strong LBP 812. MemGuard is a credible unconstrained baseline (LiRA 0.813 $\rightarrow$ 0.667) but ExactFrac -infeasible; HARP is

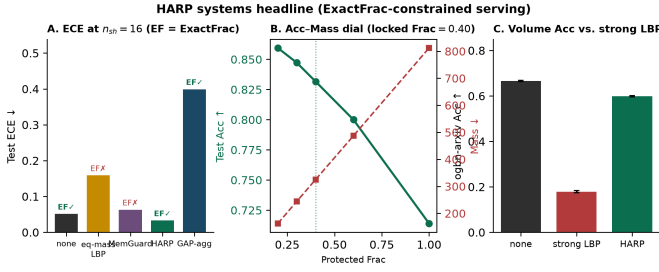


Fig. 3. Systems headline. (A) ECE at $n_{sh}=16$ with ExactFrac feasibility marks. (B) Frac Acc-Mass dial. (C) ogbn-arxiv Acc vs. strong LBP.

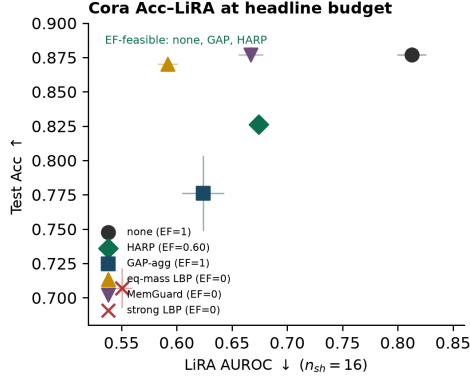


Fig. 4. Cora Acc-LiRA at the headline budget ($n_{sh}=16$, 5 seeds). ExactFrac-feasible: none, GAP-agg, HARP. Equal-mass LBP/MemGuard win unconstrained Acc-LiRA but ExactFrac=0.

TABLE II
CORA GRAPH-SAGE UNDER EXACTFRAC SLA $c=0.60$ (5 SEEDS; DEFENSE-AWARE LiRA AT $n_{sh}=16$). FEASIBLE UNDER $c>0$: NONE, GAP-AGG (TRAIN-ONLY NOISE $\Rightarrow$ EXACTFRAC=1), HARP.
†GAP-INSPIRED AGGREGATION LAPLACE WITH ANALYTICAL ε (NOT A FULL GAP CODEBASE). §SHADOW-FIT MEMGUARD; ϕ IS THE STANDARDIZED-MLP ATTACK IT OPTIMIZES AGAINST. ¶HARP AT FRAC=1 RECOVERS STRONG-LBP ACC-LiRA (UNCONSTRAINED PARITY).

Defense	Feas.	Acc	LiRA	TPR@1%	ϕ
none	yes	0.877	0.813	0.200	0.650
GAP-agg ($\sigma=3$, $\varepsilon \approx 41$) [†]	yes	0.776	0.624	0.039	0.540
HARP (release)	yes	0.826	0.674	0.060	0.626
LBP equal-mass	no	0.870	0.592	0.009	0.584
MemGuard [§]	no	0.877	0.667	0.080	0.619
LBP $b=0.3$ / HARP-full [¶]	no	0.707 / 0.715	0.551 / 0.560	0.007 / 0.008	0.565 / 0.572

within ≈ 0.01 LiRA of MemGuard with ExactFrac=0.60 and much lower ECE. At $n_{sh}=4$: GTD 0.878/0.538; MaskArmor 0.877/0.580.

GTD/MaskArmor report Acc/LiRA/ECE only (train-time; no Mass/Frac). GAP-agg is ExactFrac-feasible (train-only noise $\Rightarrow$ ExactFrac=1) with finite ε , at Acc/ECE cost. HARP is the high-Acc ExactFrac option without training DP; HARP-full recovers strong-LBP Acc-LiRA under $c=0$. Selective masking keeps ExactFrac=0.60 but raises LiRA.

d) *HARPoGAP: DP trainer as a component.*: Proposition 3 says selective release is post-processing, so composing HARP on a GAP-trained model preserves GAP's (ε, δ) . Measured (Cora, 5 seeds, $\sigma=3$): GAP alone Acc 0.770 ± 0.030 ,

TABLE III
HARP PORTABILITY GRID (GRAPHSAGE; CORA 5-SEED, OTHERS 3-SEED; $n_{sh}=4$). HEADLINE CORA AUDITS USE $n_{sh}=16$ (TABLE II). TPR = LiRA TPR@1% FPR.

Dataset	Defense	Acc	LiRA	TPR	Mass	Frac
Cora	none	0.877	0.567	0.033	—	—
	LBP $b=0.3$	0.707	0.517	0.019	812	1.00
	HARP	0.832	0.533	0.011	325	0.40
Citeseer	none	0.744	0.591	0.042	—	—
	LBP $b=0.3$	0.620	0.523	0.030	998	1.00
	HARP	0.716	0.548	0.014	399	0.40
Chameleon	none	0.448	0.618	0.059	—	—
	LBP $b=0.3$	0.391	0.539	0.038	683	1.00
	HARP	0.452	0.555	0.011	273	0.40
Actor	none	0.335	0.610	0.037	—	—
	LBP $b=0.3$	0.287	0.549	0.032	2280	1.00
	HARP	0.327	0.561	0.019	912	0.40
PubMed	none	0.876	0.505	0.012	—	—
	LBP $b=0.3$	0.768	0.499	0.010	5915	1.00
	HARP	0.839	0.502	0.012	2366	0.40
Photo	none	0.802	0.500	—	—	—
	LBP $b=0.3$	0.552	0.499	—	2295	1.00
	HARP	0.737	0.501	—	946	0.40

LiRA 0.520; HARPoGAP at Frac=0.40 keeps the same analytical $\varepsilon \approx 41$, ExactFrac=0.60, LiRA 0.517, clean-slice conf AUROC 0.584, at Acc 0.649—composition costs Acc because GAP posteriors already carry training noise. The large ε is the Acc-preserving operating point of our GAP-agg reimplementation ($\sigma=5$ yields $\varepsilon \approx 19$ at Acc 0.44); we do not market $\varepsilon \approx 41$ as a strong privacy guarantee—only as a finite analytical budget that HARP does not inflate. Operators who need a formal ε and selective-release controls can compose; operators who need only ExactFrac keep plain HARP's higher Acc.

Slice ECE localizes the win on Cora (5 seeds): $ECE_{clean}=0.018$ vs. $ECE_{prot}=0.081$ (population 0.033); clean nodes match undefended calibration, protected nodes carry the noise penalty. The pattern is portable: clean-slice ECE beats undefended population ECE on Cora/Citeseer/Chameleon (0.020/0.083/0.273 vs. 0.053/0.140/0.290), while equal-mass LBP perturbs every response even when its population ECE looks competitive (Citeseer 0.073). Δ Acc vs. strong LBP at Frac=0.40 ranges from +0.040 (Actor) to +0.183 (Photo) across six graphs at matched telemetry (MassHARP=0.4 MassLBP).

e) *Multi-dataset portability.*: Table III checks that selective release remains ExactFrac-feasible across six graphs under a matched light audit ($n_{sh}=4$; headline Cora uses $n_{sh}=16$). The secondary comparator is published strong LBP ($b=0.3$), which collapses Acc under full-graph noise; HARP recovers 4–19 Acc points at Frac=0.40 while keeping ExactFrac=0.60. This is *not* an unconstrained Acc-LiRA win (equal-mass LBP remains preferred when $c=0$). Heterophilic rows (Chameleon, Actor) show the largest undefended LiRA and the clearest case for a Frac budget; PubMed/Photo are near-chance controls where operators should set Frac=0 unless ExactFrac itself is required.

f) *Frac dial and Mass accounting.*: Table IV sweeps Frac on Cora (5 seeds): Acc falls monotonically as Frac rises

TABLE IV

HARP FRAC SWEEP ON CORA (5 SEEDS; $\sigma_{\text{strong}}=0.30$). ACC/MASS ARE BUDGET-INVARIANT; LiRA COLUMN IS $n_{\text{sh}}=4$ (SEE TABLE II/ V FOR $n_{\text{sh}}=16$).

Frac	Acc $\uparrow$	LiRA $\downarrow$	Mass $\downarrow$
0.20	0.859 [0.855, 0.866]	0.537 [0.532, 0.543]	162
0.30	0.847 [0.843, 0.853]	0.538 [0.533, 0.543]	244
0.40	0.832 [0.826, 0.839]	0.533 [0.525, 0.541]	325
0.60	0.800 [0.793, 0.809]	0.519 [0.512, 0.526]	487
1.00	0.714 [0.699, 0.727]	0.511 [0.503, 0.519]	812

TABLE V

CFS ON CORA AT $n_{\text{sh}}=16$ UNDER EXACTFRAC ≥ 0.60 AND LiRA $\leq \tau=0.70$ (3 SEEDS; MEAN SELECTED FRAC/ACC/LiRA). ALL ROWS FEASIBLE_AUDIT=1.

Constructor	Selected Frac	Acc	LiRA
topology (LTE)	0.33	0.838	0.683
ensemble	0.30	0.838	0.681
random	0.23	0.839	0.689

under fixed σ_{strong} —the systems Acc–Mass dial. LiRA in this sweep uses $n_{\text{sh}}=4$ (cheap dial diagnostics); the headline privacy comparison is Table II at $n_{\text{sh}}=16$, and CFS under that budget is Table V.

g) *Constrained Frac Search (matched to the headline audit).*: CFS is only meaningful when τ tracks the attack budget: $\tau=0.55$ is appropriate under $n_{\text{sh}}=4$ but infeasible under headline $n_{\text{sh}}=16$ (locked HARP LiRA ≈ 0.67). We therefore run CFS at $n_{\text{sh}}=16$ with ExactFrac ≥ 0.60 and modern cap $\tau=0.70$ (Table V). Among Frac $\in \{0, 0.2, 0.3, 0.4\}$, every constructor finds a feasible point; mean selected Frac is 0.23–0.33 with Acc ≈ 0.84 —strictly better Acc than the locked Frac=0.40 dial while meeting τ . Tighter $\tau=0.68$ pushes topology to Frac=0.40; looser $\tau=0.75$ selects Frac=0.20. Under ExactFrac alone (no τ), CFS degenerates to Frac=0; the audit cap is what forces a nonzero spend. Locked Frac=0.40 remains the conservative fixed operating point.

h) *Exact scores $\neq$ membership safety.*: Clean nodes are well calibrated (ECE $_{\text{clean}}=0.018$), but they are not automatically hard to attack. Under locked HARP, unprotected-slice confidence AUROC is ≈ 0.64 vs. ≈ 0.51 population-wide (Table XI); the mitigation stack below addresses it. Aerni et al. [37] motivate such canary checks (Sec. VI, canaries).

i) *Velocity: multi-query and session budgets.*: Averaging K independent releases denoises Laplace and moves toward the undefended baseline (Cora 5-seed LiRA at $K=1/5/20$: HARP 0.536/0.546/0.551; $n_{\text{sh}}=4$). A session policy of at most B fresh draws per *authenticated principal*—reusing the last draw thereafter—caps this drift (Table VI). **Enforcement assumption:** server-side state bound to authenticated identities with rate limits. Sybil-capable attackers who mint many principals can bypass per-identity B ; Table VI shows $N=5$ identities under per-id $B=1$ recover Acc ≈ 0.874 (near uncapped), while a *shared global* $B=1$ pool keeps Acc at the one-shot floor (0.825). Deploy shared B (or authenticated single-principal sessions) when Sybil velocity is in scope; Frac and B remain

TABLE VI

SESSION BUDGET ON HARP CORA AT $K=20$ (5 SEEDS). $B=1$ FREEZES ACC NEAR ONE-SHOT; $B=20$ MATCHES UNCAPPED. SHARED GLOBAL $B=1$ ACROSS N IDENTITIES BLOCKS SYBIL ACC RECOVERY (RIGHT).

Policy	B	Acc	LiRA	N ids	Per-id $B=1$	Shared $B=1$
Budget reuse	1	0.832	0.540	1	0.825	0.825
Budget reuse	5	0.851	0.540	5	0.874	0.825
Budget reuse	20	0.885	0.545	20	0.876	0.825
Uncapped avg	—	0.886	0.545	—	—	—

TABLE VII

RELEASE-ONLY SELECTIVE CONSTRUCTORS AT FRAC=0.40 / EXACTFRAC=0.60 (CORA, 5 SEEDS; MEAN ± 1.96 SE). SLICE = CLEAN-SLICE CONF AUROC (ATTACK-BUDGET FREE). POPULATION LiRA HERE IS $n_{\text{sh}}=4$; HEADLINE LiRA IS TABLE II AT $n_{\text{sh}}=16$.

Policy	Feas. $c=0.6$	Acc	LiRA	Slice $\downarrow$	ExactFrac
none	yes	0.877 ± 0.005	0.567 ± 0.006	—	1.00
GAP-agg ($\sigma=3$)	yes	0.776 ± 0.024	0.518 ± 0.008	—	1.00
LBP equal-mass	no	0.870 ± 0.005	0.520 ± 0.011	—	0.00
LBP $b=0.3$ / HARP-full	no	0.707 / 0.715	0.517 / 0.512	—	0.00
sel. audit	yes	0.820 ± 0.010	0.526 ± 0.009	0.616 ± 0.016	0.60
sel. ensemble	yes	0.817 ± 0.011	0.527 ± 0.014	0.594± 0.010	0.60
sel. random	yes	0.810 ± 0.006	0.526 ± 0.007	0.669 ± 0.015	0.60
sel. entropy	yes	0.805 ± 0.010	0.530 ± 0.014	—	0.60
sel. topology (LTE)	yes	0.826 ± 0.006	0.536 ± 0.008	0.647 ± 0.012	0.60

complementary controls.

j) *Constructors: a privacy effect, not just an interface.*: The decisive constructor metric is *clean-slice* conf AUROC: ranking removes vulnerable members from the bit-exact slice (Table VII). At matched ExactFrac=0.60 (5-seed Cora), slice AUROC falls with ranking quality: random 0.669 ± 0.015 , topology 0.647 ± 0.012 , audit 0.616 ± 0.016 , ensemble 0.594 ± 0.010 . Paired tests vs. random: audit $\Delta=0.053$ ($p<10^{-3}$), ensemble $\Delta=0.075$ ($p<10^{-3}$); Acc is equal-or-better (ensemble 0.813 vs. random 0.811). At the headline budget ($n_{\text{sh}}=16$, Frac=0.40), ensemble also lowers population LiRA vs. topology (0.646 vs. 0.674)—constructors are primarily a slice dial, with a measurable population effect under strong audits. Hop $k\in\{0, 1, 2\}$ at matched Frac changes Acc by $\lesssim 0.01$ (release-only); selectivity, not train-time alignment, carries ExactFrac.

k) *Why ExactFrac matters (serving evidence).*: Prediction caches assume deterministic outputs [40], [41]. Table VIII separates ExactFrac from sticky noise on Cora (3 seeds; 8 clients; 20k Zipf queries; shared LRU 1024): sticky LBP restores *some* shared-cache hits (0.62) but cross-client ExactFrac and pre-session audit replay stay ≈ 0 , and clean-slice threshold decisions still flicker (≈ 0.09). HARP keeps cross-client ExactFrac=0.60, zero clean-slice flicker, audit replay 0.60, and higher shared-cache hits (0.77). Table X confirms bitwise re-query rates; Table IX reports single-stream QPS. This is in-process mechanism evidence, not a production-trace study.

l) *Clean-slice residual risk (first-class) and the mitigation stack.*: Unprotected nodes are *not* claimed safe (Table XI): clean-slice conf AUROC rises to 0.63–0.67 while population LiRA stays ≈ 0.53 . HARP attacks this from three composable

TABLE VIII

STICKY NOISE $\neq$ EXACTFRAC (CORa, 3 SEEDS). CROSS-CLIENT EXACTFRAC / PRE-SESSION AUDIT REPLAY / CLEAN-SLICE THRESHOLD FLICKER / SHARED MULTI-CLIENT CACHE HIT. STICKY LBP FREEZES ONE DRAW PER CLIENT; HARP KEEPS A SHARED CLEAN MAJORITY.

Policy	Cross-EF $\uparrow$	Audit replay $\uparrow$	Clean flick. $\downarrow$	Shared hit $\uparrow$
none	1.000	1.000	0.000	0.869
LBP fresh	0.000	0.000	0.089	0.004
LBP sticky ($B=1$)	0.000	0.000	0.089	0.619
HARP (Frac 0.40)	0.600	0.600	0.000	0.769

TABLE IX

IN-PROCESS SINGLE-STREAM SERVING BENCH (CORa; 20K QUERIES). CONTENT-ADDRESSABLE KEYS ON BIT-EXACT POSTERIORIS.

Workload	Policy	Hit	QPS	p50 ms	p95 ms
zipf	none	0.937	4.0×10^6	<0.001	<0.001
zipf	lbp_fresh	0.000	9.3×10^3	0.104	0.116
zipf	harp_fresh	0.606	6.2×10^5	<0.001	0.004
zipf	lbp_B1	0.937	1.5×10^5	<0.001	0.101
zipf	harp_B1	0.937	2.2×10^6	<0.001	<0.001

TABLE X

MEASURED (NOT DEFINITIONAL) REPLAY BEHAVIOR ACROSS INDEPENDENT RE-QUERIES (CORa, 3 SEEDS): BITWISE-EQUAL FRACTION, TOP-1 LABEL FLIPS, AND THRESHOLD ($p_{\text{top}} > 0.5$) ROUTE FLIPS. HARP'S FLIPS ARE CONFINED TO THE PROTECTED MINORITY; THE CLEAN 60% IS BIT-STABLE WITH ZERO FLICKER.

Policy	Bit-equal re-query	Top-1 flip	Thresh. flip
none	1.000	0.000	0.000
GAP-agg	1.000	0.000	0.000
LBP eq-mass	0.000	0.027	0.101
HARP (Frac 0.40)	0.600	0.146*	0.181*

*Entirely within the protected 40% ($\sigma=0.30$); clean-slice flip rate is exactly 0. LBP spreads lower-amplitude flicker over *all* nodes, so no response is replay-stable.

directions (all at ExactFrac=0.60, 5 seeds): (i) *constructor*: ensemble ranking removes vulnerable members from the exact slice, $0.669 \rightarrow 0.594$ (Table VII); (ii) *deterministic confidence smoothing (DCS)*: released rows on the clean slice with $p_{\text{max}} > \theta$ are flattened by a fixed temperature ($\theta=0.9$, $T=3$)—a deterministic, argmax-preserving post-map, so re-queries stay bitwise stable, caches and audit replay still work, and routing decisions are unchanged; slice AUROC falls $0.650 \rightarrow 0.618$ under topology ranking and $0.591 \rightarrow 0.576$ on top of ensemble, with *zero* Acc change; (iii) *slice-aware CFS*: adds a slice-AUROC cap to the search. DCS trades raw-score fidelity on smoothed rows for slice privacy while preserving every operational property that motivates ExactFrac (determinism, replay, thresholds); when bit-exactness to the raw posterior is contractual, skip DCS. The stacked floor (≈ 0.575) is structural at $c=0.60$: a 60% deterministic slice must retain some membership signal. Lower c , or use GAP-agg / HARP>GAP, if tighter slice control is required.

m) *Inductive edge exclusion*: Dropping test-incident edges (CorA, 5 seeds) raises undefended LiRA to 0.818; HARP recovers Acc $0.476 \rightarrow 0.601$ vs. strong LBP at Frac=0.40.

TABLE XI

LTE-AWARE ADAPTIVE QUERIES ON CORa HARP (5 SEEDS; CONF AUROC IS ATTACK-BUDGET FREE; LiRA@0.4 USES $n_{\text{sh}}=4$).

Query set	Conf@0.4	Conf@0.6	LiRA@0.4
Population	0.518	0.473	0.533
Unprotected (clean)	0.645	0.633	0.542
Boundary (clean)	0.634	0.628	0.538
Top-decile LTE	0.641	0.511	0.529

TABLE XII

AUDIT CONTROLS (GRAPHSAGE; 3 SEEDS). *HETEROPHILIC. (b) COMPARATIVE SHADOW SCALING (NOT HARP-ONLY).

(a) Undefended LiRA ($n_{\text{sh}}=4$)			(b) Cora shadow scaling (LiRA / TPR@1%)			
Dataset	n	Acc	LiRA	none	LBP	HARP
Chameleon*	2 277	0.449	0.623	0.568 / 0.033	0.517 / 0.022	0.534 / 0.014
Cora	2 708	0.879	0.668	0.808 / 0.201	0.555 / 0.008	0.670 / 0.057
Actor*	7 600	0.335	0.610	0.847 / 0.265	0.561 / 0.008	0.677 / 0.075
Photo	7 650	0.825	0.500	Acc-invariant: none 0.879, LBP 0.716, HARP 0.505 0.836. Chameleon mirrors the pattern (at $n_{\text{sh}}=64$: none 0.965, LBP 0.685, HARP 0.811; Δ Acc vs. LBP +0.061).		
PubMed	19 717	0.876	0.505			
ogbn-arxiv	169 343	0.666	0.505			

C. When to spend: volume $\times$ structure

Table XII(a): undefended LiRA concentrates in small/mid heterophilic cells; Photo/PubMed/ogbn-arxiv sit near chance under matched shadows—audit first, then set Frac. Table XII(b) reports the honest shadow-count story on Cora for none / LBP / HARP. Absolute LiRA rises with n_{shadows} for every defense [4]; Acc is invariant. HARP does *not* beat strong LBP on LiRA at large shadow counts (Cora $n_{\text{sh}}=64$: LBP 0.561 vs. HARP 0.677)—expected when ExactFrac=0 is allowed. What HARP keeps is (i) strictly lower LiRA than undefended and (ii) Acc recovery vs. strong LBP (Δ Acc +0.12, invariant in n_{sh}). TPR@1% FPR follows the same pattern.

D. Canaries and large-graph volume

Cora canary checks: HARP suppresses planted/top-decile confidence cues relative to none/LBP (top-decile conf AUROC 0.551/0.548/0.266 for none/LBP/HARP; planted 0.527/0.543/0.423); population LiRA remains primary. Table XIV: on ogbn-arxiv ($\sim 169k$; NeighborLoader; gate off, warmup 3) [28], strong LBP drops Acc to 0.179 [0.175, 0.184]. HARP recovers 0.599 [0.597, 0.601] (paired Δ Acc +0.420 [0.414, 0.425]; 3 seeds) at Frac=0.40. Undefended LiRA stays near chance at $n_{\text{sh}} \in \{2, 4\}$ and also at $n_{\text{sh}}=16$ (LiRA 0.510; Acc 0.666)—so the Acc recovery is a serving result under an audit-null regime, not under-powered shadows. On ogbn-products ($\sim 2.4M$ nodes) [28], Table XIII reports a serving probe plus BFS LiRA at $n \in \{15k, 40k\}$ and $n_{\text{sh}} \in \{2, 4, 8\}$. Full-graph products LiRA OOMs on this host; we do *not* claim a million-node privacy audit. Across standard-recipe probes LiRA stays near chance while HARP recovers Acc vs. strong LBP (e.g., $15k/n_{\text{sh}}=8$: HARP Acc 0.693 vs. LBP 0.343).

a) *Canary-stressed probe (above-chance signal at scale)*: Standard products/arxiv recipes sit near chance, so Acc recovery alone is a weak privacy claim. We plant 300

TABLE XIII

OGBN-PRODUCTS: STANDARD BFS LADDER (NEAR-CHANCE) VS. CANARY-STRESSED 15K PROBE ($n_{sh}=4$, 3 SEEDS). CAN. = CANARY-ONLY LiRA.

Setting	Acc	LiRA	Can.
Full load none (2.45M)	0.695	—	—
Std. 15k none/LBP/HARP	0.876/0.343/0.693	0.528/0.501/0.513	—
Canary none	0.861	0.593	0.997
Canary LBP $b=0.3$	0.379	0.581	0.805
Canary HARP	0.685	0.592	0.891

TABLE XIV

OGBN-ARXIV VOLUME (3 SEEDS). LiRA_{2/4} ARE DEFENSE-AWARE; UNDEFENDED LiRA AT $n_{sh}=16$ IS 0.510 (NEAR-CHANCE CREDIBILITY CHECK).

Def.	Acc	LiRA ₂	LiRA ₄	Mass
none	0.666	0.502	0.503	—
LBP	0.179	0.498	0.501	50 803
HARP	0.599	0.514	0.505	20 322

unique-feature canaries on the 15k BFS probe (half in train / half held out; $n_{sh}=4$; 3 seeds): undefended canary-LiRA is ≈ 1.00 while population LiRA is 0.593 (Table XIII). HARP cuts canary-LiRA to 0.891 (still high) at Acc 0.685 with ExactFrac=0.60; strong LBP reaches 0.805 only by collapsing Acc to 0.379 at ExactFrac=0. Population LiRA barely moves (≈ 0.59) under either release defense—defense-aware shadows recalibrate—so the scale takeaway is Acc-preserving ExactFrac under a hard canary stress, not a population-LiRA win at 15k. We do *not* claim a full 2.4M-node privacy audit.

If audit LiRA ≈ 0.5 , prefer Frac=0; if LiRA $\gg 0.5$ or ExactFrac $\geq c$ is required, set Frac on val Acc.

b) Operator workflow.: Audit with defense-aware LiRA ($n_{sh}\geq 16$ when affordable); if LiRA ≈ 0.5 set Frac=0; else CFS on val Acc under (c, τ) (Table IV); prefer ensemble constructors + optional DCS for slice risk; enforce shared $B=1$ under Sybil velocity; use HARPoGAP when a legal ε is required.

VII. DISCUSSION

HARP is for ExactFrac $\geq c>0$ with usable Acc—not for beating uniform noise when $c=0$ is allowed. ExactFrac formalizes a serving assumption already present in prediction caches [40], [41]; $c=0.60$ is a study default, and sticky noise is not a substitute (Table VIII). GAP-agg already meets ExactFrac=1 with finite (often large) ε ; HARP’s complementary claim is high-Acc ExactFrac without training DP, plus constructors/CFS/slice controls (Prop. 3). **Limits:** (i) ExactFrac $>0 \Rightarrow$ infinite global ε on released scores; (ii) clean-slice AUROC stays above chance at $c=0.60$ —structural; (iii) CFS τ must track the LiRA budget; (iv) products privacy is a BFS/canary ladder; (v) GAP-agg/MemGuard are reimplementations; (vi) serving benches are in-process, not production traces. Future: learned constructors, distributed full-graph LiRA, production API traces.

VIII. CONCLUSION

HARP is an ExactFrac-constrained score-serving framework for GNN APIs: selective release meets $c>0$ where uniform noise cannot, constructors cut clean-slice attack AUROC with significance, CFS is grid-optimal, and HARPoGAP preserves training DP. Headline evidence uses $n_{sh}=16$; residual clean-slice risk and scoped large-graph audits are reported honestly. Use training-time DP (alone or under HARP) when a legal global ε is required.

a) Reproducibility.: Code (defenses/harp.py, run_nsh16_headline.py, run_cfs_nsh16.py, run_exactfrac_sla_evidence.py, run_products_canary_leakage.py, run_bulletproof.py), locked configs, and frozen CSVs: <https://github.com/krishharma/privacy-gnn>. Headline LiRA $n_{shadows}=16$; CFS uses $\tau=0.70$; ExactFrac SLA evidence in Table VIII.

REFERENCES

- [1] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in *Proc. IEEE Symp. Security Privacy (S&P)*, 2017, pp. 3–18.
- [2] A. Salem, Y. Zhang, M. Humbert, P. Berrang, M. Fritz, and M. Backes, “ML-Leaks: Model and data independent membership inference attacks and defenses on machine learning models,” in *Proc. NDSS*, 2019.
- [3] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, “Privacy risk in machine learning: Analyzing the connection to overfitting,” in *Proc. IEEE CSF*, 2018, pp. 268–282.
- [4] N. Carlini, S. Chien, M. Nasr, A. Song, T. Terzis, and F. Tramer, “Membership inference attacks from first principles,” in *Proc. IEEE S&P*, 2022, pp. 1897–1914.
- [5] I. E. Olatunji, W. Nejdl, and M. Khosla, “Membership inference attack on graph neural networks,” arXiv:2101.06570, 2021.
- [6] J. Jia, A. Salem, M. Backes, Y. Zhang, and N. Z. Gong, “MemGuard: Defending against black-box membership inference attacks via adversarial examples,” in *Proc. ACM CCS*, 2019, pp. 259–274.
- [7] Y. Wu, D. Li, B. Chen, and L. Shen, “Adapting membership inference attacks to GNN for graph classification,” in *Proc. IEEE ICDM*, 2021, pp. 1429–1434.
- [8] Y. Zhang, Y. Zhao, Z. Li, X. Cheng, Y. Wang, O. Kotevska, P. S. Yu, and T. Derr, “A survey on privacy in graph neural networks: Attacks, preservation, and applications,” arXiv:2308.16375, 2023.
- [9] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. ICLR*, 2017.
- [10] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. NeurIPS*, 2017.
- [11] Y. Rong, W. Huang, T. Xu, and J. Huang, “DropEdge: Towards deep graph convolutional networks on node classification,” in *Proc. ICLR*, 2020.
- [12] C. A. Choquette-Choo, F. Tramer, N. Carlini, and N. Papernot, “Label-only membership inference attacks,” in *Proc. ICML*, 2021, pp. 1964–1974.
- [13] S. Sajadmanesh et al., “GAP: Differentially private graph neural networks with aggregation perturbation,” in *Proc. USENIX Security*, 2023, pp. 3229–3246.
- [14] M. Abadi et al., “Deep learning with differential privacy,” in *Proc. ACM CCS*, 2016, pp. 308–318.
- [15] M. Nasr, R. Shokri, and A. Houmansadr, “Machine learning with membership privacy using adversarial regularization,” in *Proc. ACM CCS*, 2018, pp. 634–646.
- [16] M. Nasr, R. Shokri, and A. Houmansadr, “Comprehensive privacy analysis of deep learning,” in *Proc. IEEE S&P*, 2019, pp. 739–753.
- [17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017, pp. 1321–1330.
- [18] J. Zhu et al., “Beyond homophily in graph neural networks: Current limitations and effective designs,” in *Proc. NeurIPS*, 2020.
- [19] H. Pei, B. Wei, K. C.-C. Chang, Y. Lei, and B. Yang, “Geom-GCN: Geometric graph convolutional networks,” in *Proc. ICLR*, 2020.

- [20] D. Lim, F. Hohne, X. Li, S. L. Huang, V. Gupta, O. Bhalerao, and S. N. Lim, "Large scale learning on non-homophilous graphs: New benchmarks and strong simple methods," in *Proc. NeurIPS*, 2021.
- [21] P. Niu, C. Pan, S. Chen, and O. Milenković, "Graph Transductive Defense: a Two-Stage Defense for Graph Membership Inference Attacks," arXiv:2406.07917, 2024.
- [22] C. Chen, X. Zhang, H. Qiu, J. Lou, Z. Liu, and X. Chen, "MaskArmor: Confidence masking-based defense mechanism for GNN against MIA," *Information Sciences*, vol. 669, 2024.
- [23] M. A. P. Chamikara et al., "Can diffusion-based posterior smoothing secure GNNs from membership inference attacks?" in *Proc. AICCSA*, 2025.
- [24] V. Duddu, A. Boutet, and V. Shejwalkar, "Quantifying privacy leakage in graph embedding," in *Proc. WSDM*, 2020, pp. 76–84.
- [25] M. Fredrikson, S. Jha, and T. Ristenpart, "Model inversion attacks that exploit confidence information," in *Proc. ACM CCS*, 2015, pp. 1322–1333.
- [26] F. Tramèr et al., "Stealing machine learning models via prediction APIs," in *Proc. USENIX Security*, 2016, pp. 601–618.
- [27] C. Dwork, "Differential privacy," in *Proc. ICALP*, 2006, pp. 1–12.
- [28] W. Hu et al., "Open graph benchmark: Datasets for machine learning on graphs," in *Proc. NeurIPS*, 2020.
- [29] P. Veličković et al., "Graph attention networks," in *Proc. ICLR*, 2018.
- [30] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *Proc. ICLR*, 2019.
- [31] I. Mironov, "Rényi differential privacy," in *Proc. IEEE CSF*, 2017, pp. 263–275.
- [32] B. Jayaraman and D. Evans, "Evaluating differentially private machine learning in practice," in *Proc. USENIX Security*, 2019, pp. 1895–1912.
- [33] C. Song, T. Ristenpart, and V. Shmatikov, "Machine learning models that remember too much," in *Proc. ACM CCS*, 2017, pp. 587–601.
- [34] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proc. ICML*, 2005, pp. 625–632.
- [35] Y. Jiang et al., "Unveiling privacy vulnerabilities: Investigating the role of structure in graph data," in *Proc. ACM KDD*, 2024. (GPS: structure→attribute leakage; GHRatio; learnable edge sampling.)
- [36] Q. Yuan et al., "Node-Aware Differentially Private Graph Neural Networks for Graph Representation Learning," arXiv:2308.04943, 2023.
- [37] M. Aerni, J. Zhang, and F. Tramèr, "Evaluations of machine learning privacy defenses are misleading," in *Proc. ACM CCS*, 2024.
- [38] X. He, R. Wen, Y. Wu, M. Backes, Y. Shen, and Y. Zhang, "Node-level membership inference attacks against graph neural networks," arXiv:2102.05429, 2021.
- [39] M. Khosla, "Impact of graph structure on membership-inference risk for graph neural networks," *Proc. Priv. Enhancing Technol. (PoPETs)*, 2026.
- [40] D. Crankshaw, X. Wang, G. Zhou, M. J. Franklin, J. E. Gonzalez, and I. Stoica, "Clipper: A low-latency online prediction serving system," in *Proc. USENIX NSDI*, 2017, pp. 613–627.
- [41] NVIDIA Corporation, "Triton Inference Server: Response cache," User Guide, 2024. [Online]. Available: https://docs.nvidia.com/deeplearning/triton-inference-server/user-guide/docs/user_guide/response_cache.html